

ECEN 607: Advanced Analog Circuit Tech

Lab 6:

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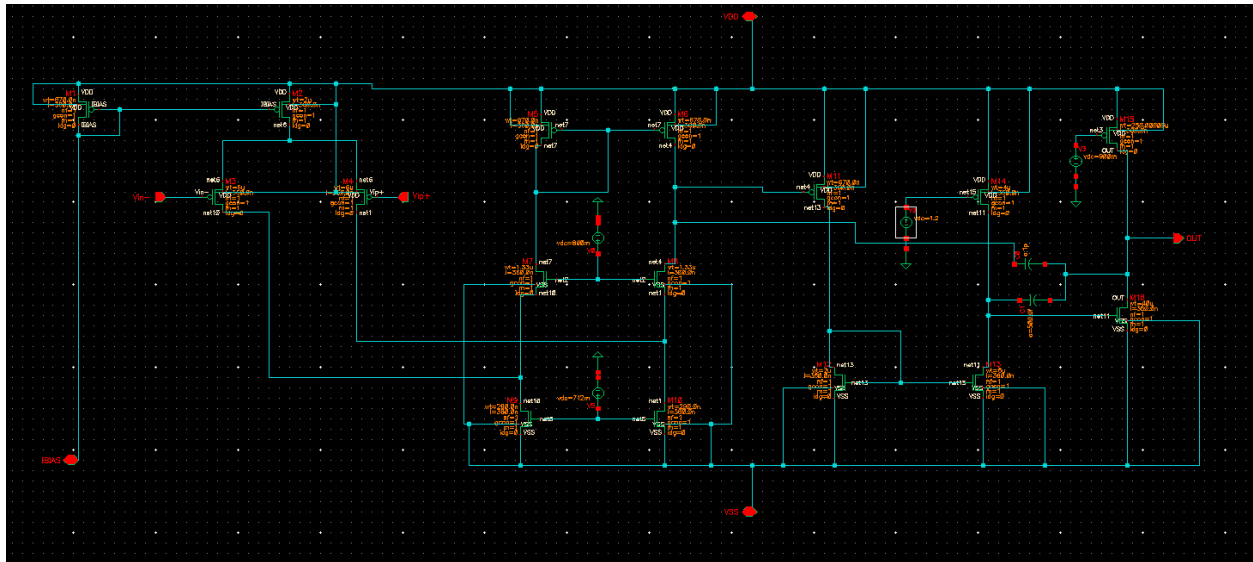
UIN: 537000654

Figure7-1: Nested Miller Opamp

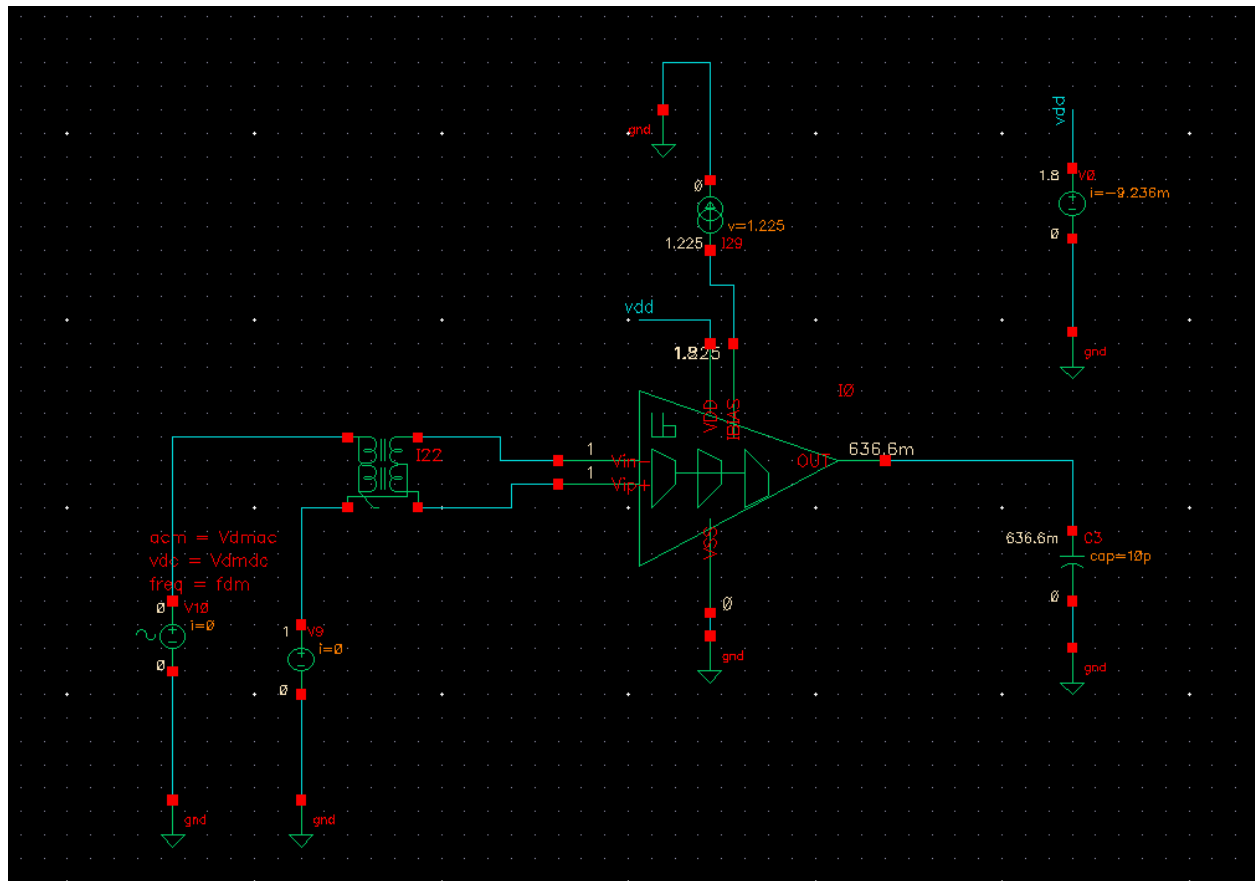
1. Simulate the circuit from the prelab (the nested miller op amp) and adjust the transistor sizes accordingly until you meet all specifications. Provide the necessary plots such that you show clearly

that you meet all the specs. Measure the input offset voltage.

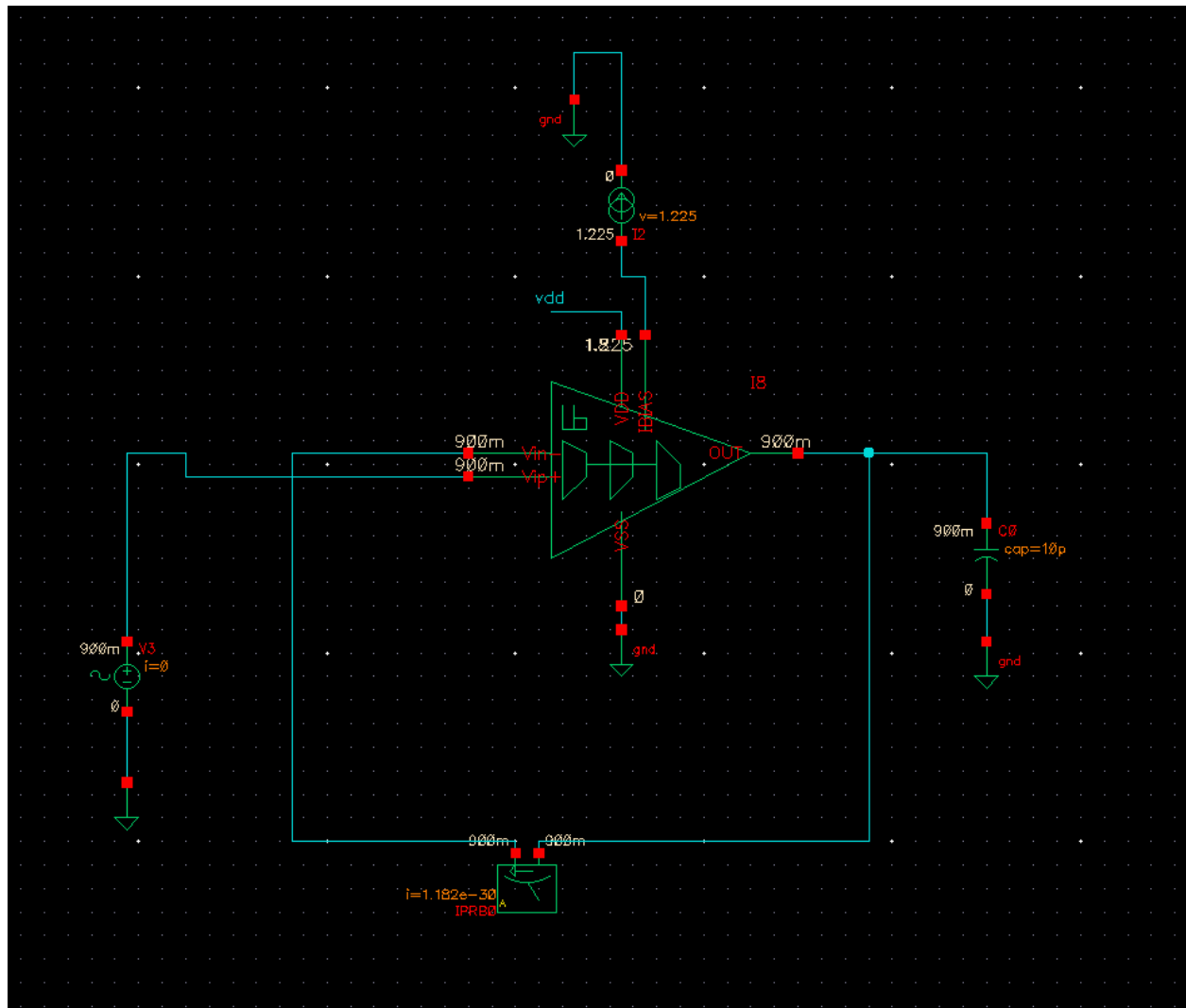
Schematic:



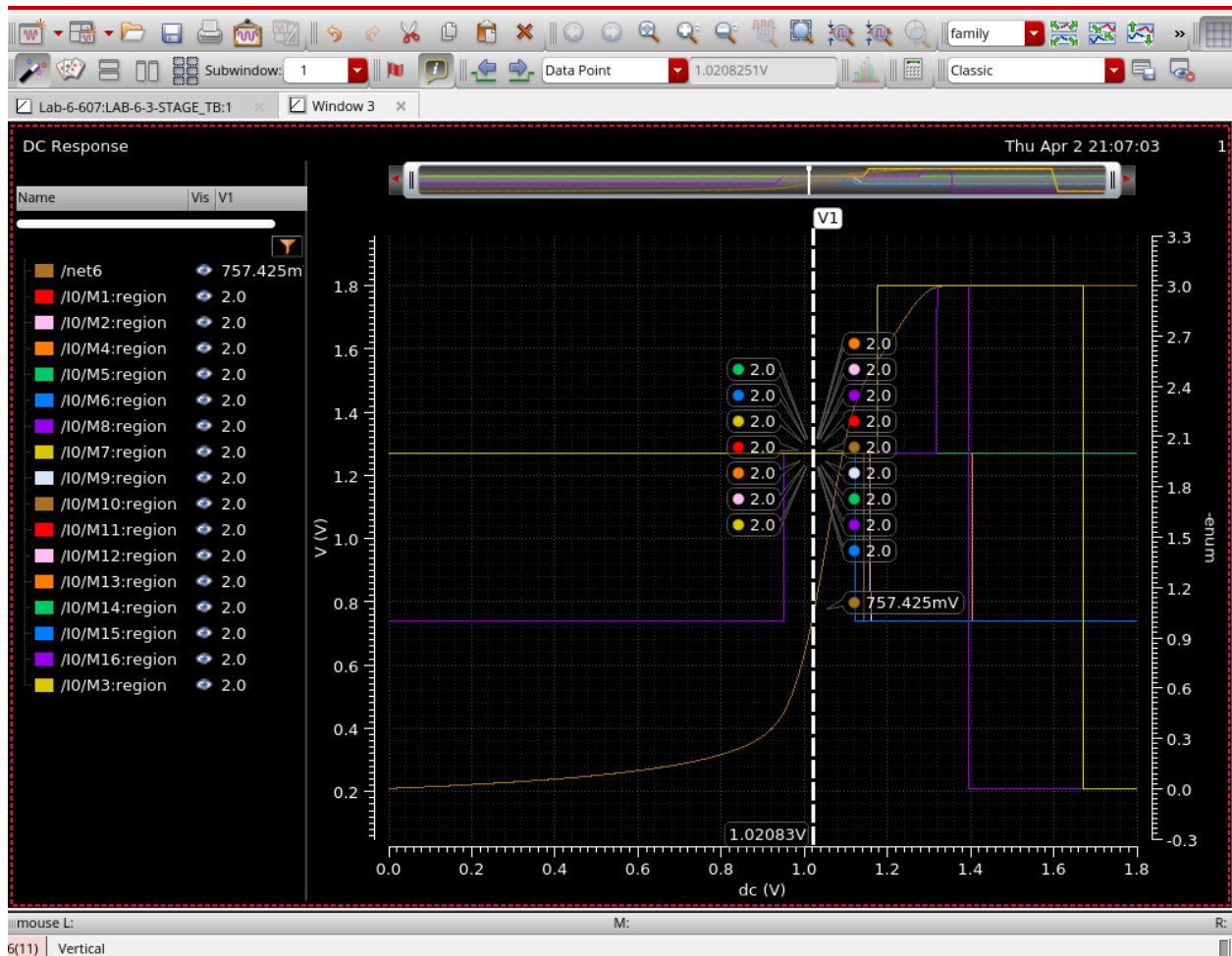
The open loop test bench set-up:



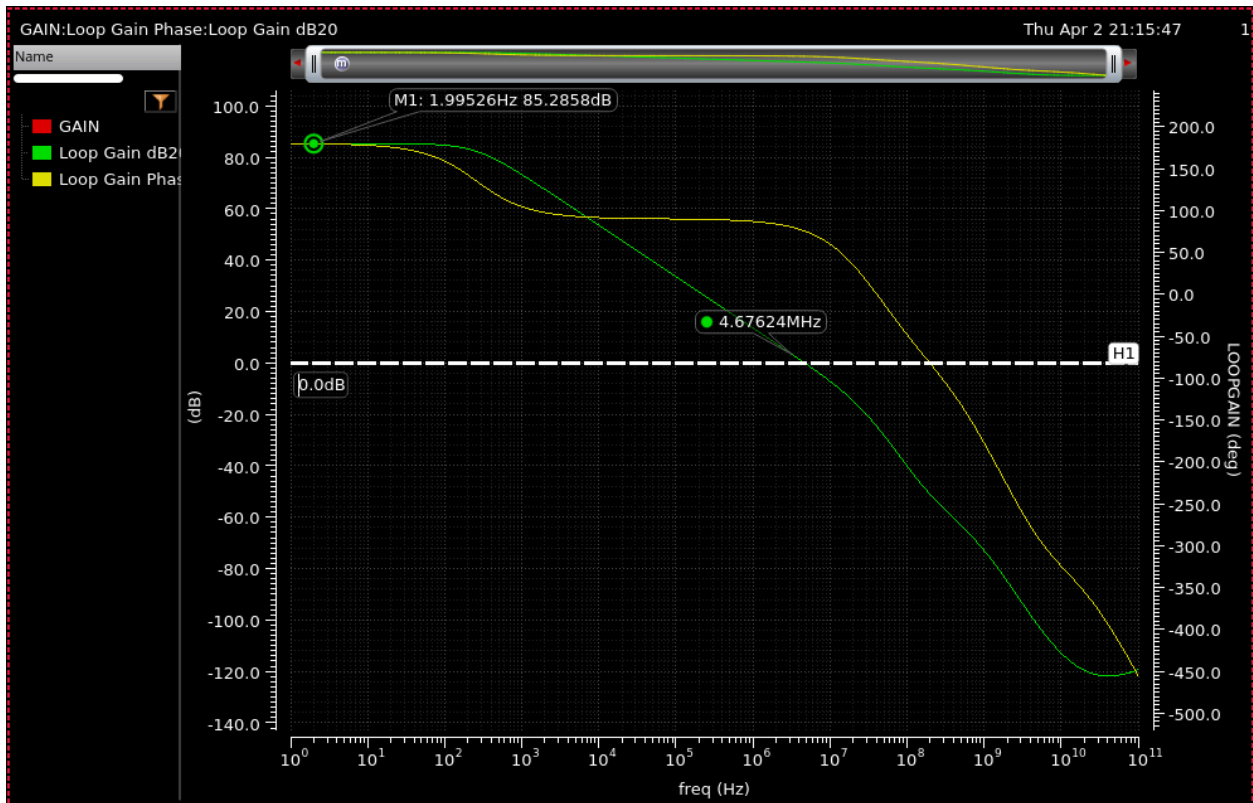
The closed loop test bench setup:



DC Analysis and bringing transistors to saturation:



AC Analysis and measuring gain:



Phase Margin and Gain Margin :

ADE XL Test Editor - Lab-6-607:LAB-6-3-STAGE_TB:1@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

	Name	Value
1	VDD	1.8
2	Vcmddc	1
3	Vcmac	1
4	Vdmddc	0
5	Vdmac	1
6	Vdmtr	0
7	fdm	0
8	id2	1u
9	CI	10p
10	p5vonly	0

Analyses

Type	Enable	Arguments
1 dc	☒	t 0 1.8 1m Linear Step Size Start-Stop /V9
2 ac	☒	1 100G 20 Logarithmic Points Per Decade Start-Stop
3 stb	☒	1 100G 20 /IPRB0 Logarithmic Points Per Decade Sta...

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Opti
3 Loop Gain Phase	wave	☒	☐	
4 Loop Gain dB20	wave	☒	☐	
5 net1		☐	☒	allv
6 Gain Margin	24.73	☒	☒	
7 Phase Margin Frequency	4.669M	☒	☐	
8 Phase Margin	75.54	☒	☐	

> Results in ...-STAGE_TB:1/simulation/LAB-6-3-S

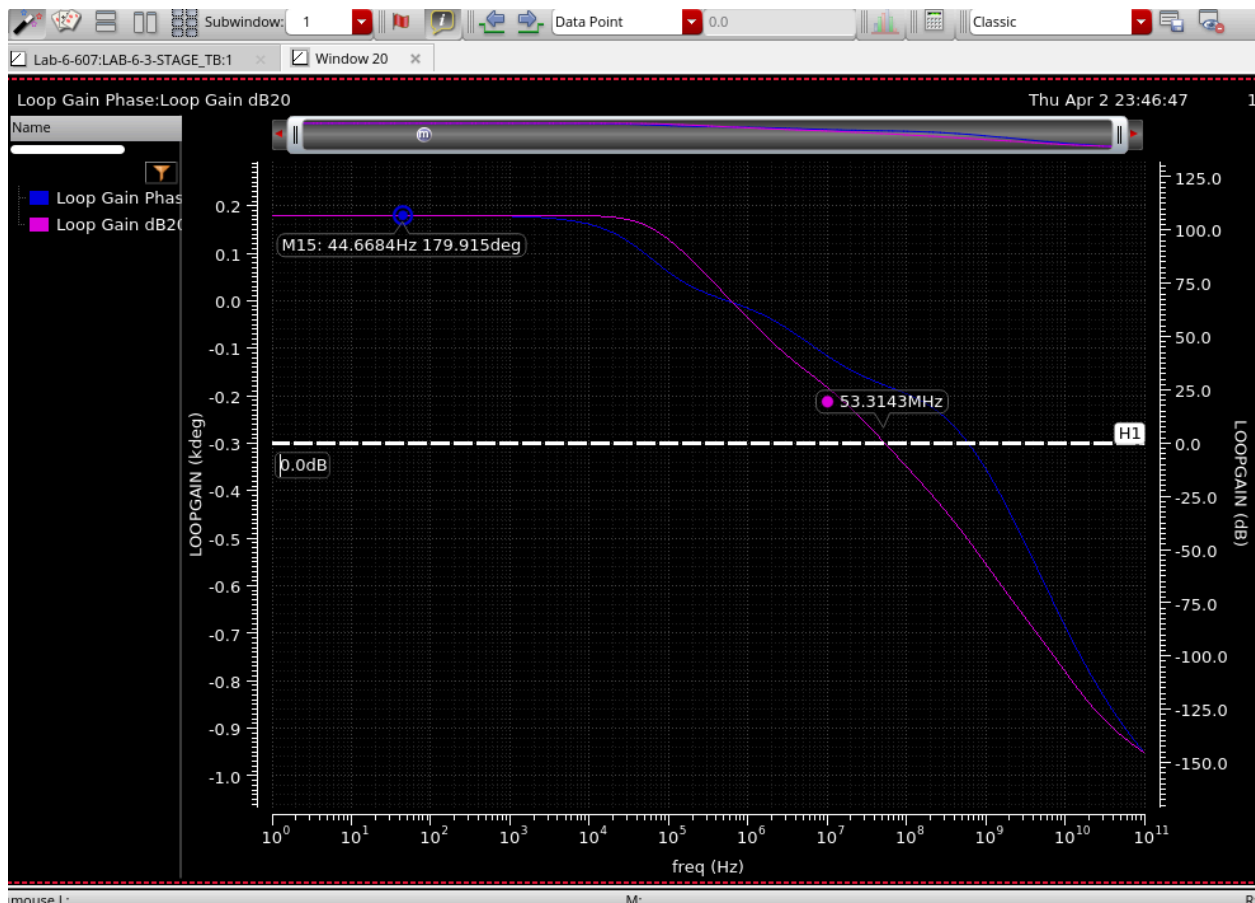
Outputs Save Operating Points

4(5) Debug Test Status: Ready T=27 C Simulator: spectre aps State: Lab-6-607:LAB-6-3-STAGE_TB:1_active

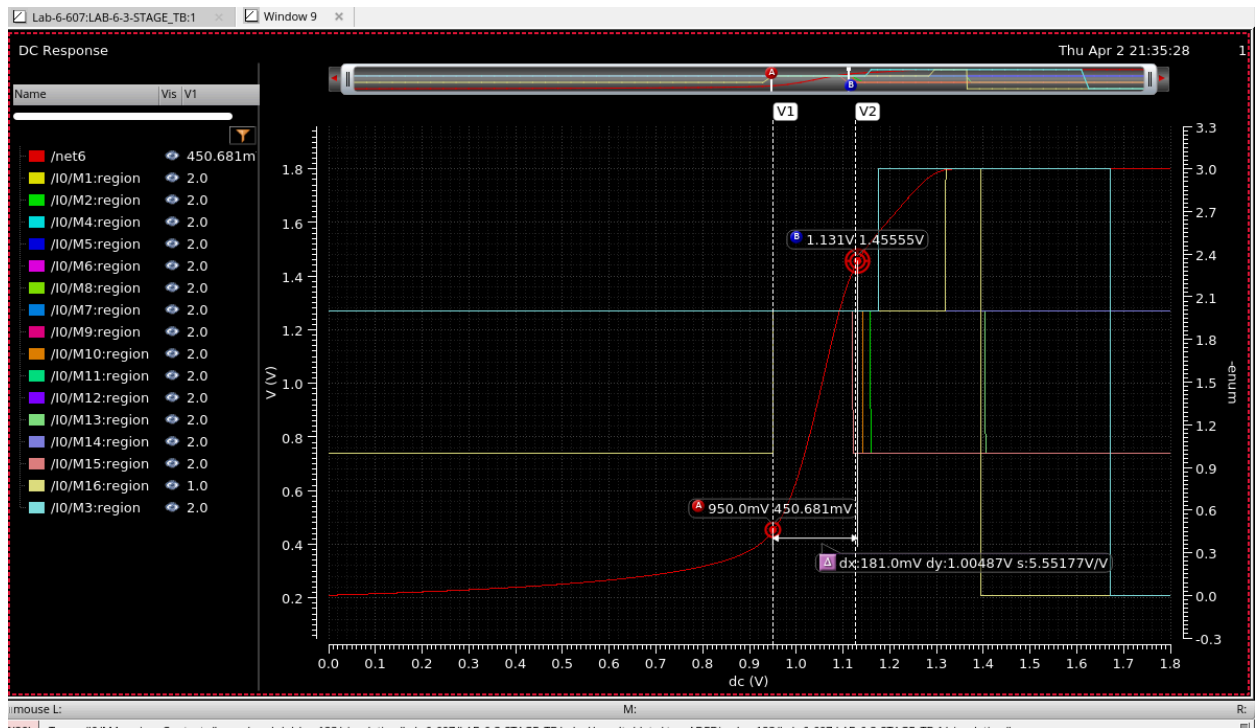
Since the gain specification is not met ,i approached it with an another design of adding one more stage to meet the gain spec:



Closed loop:



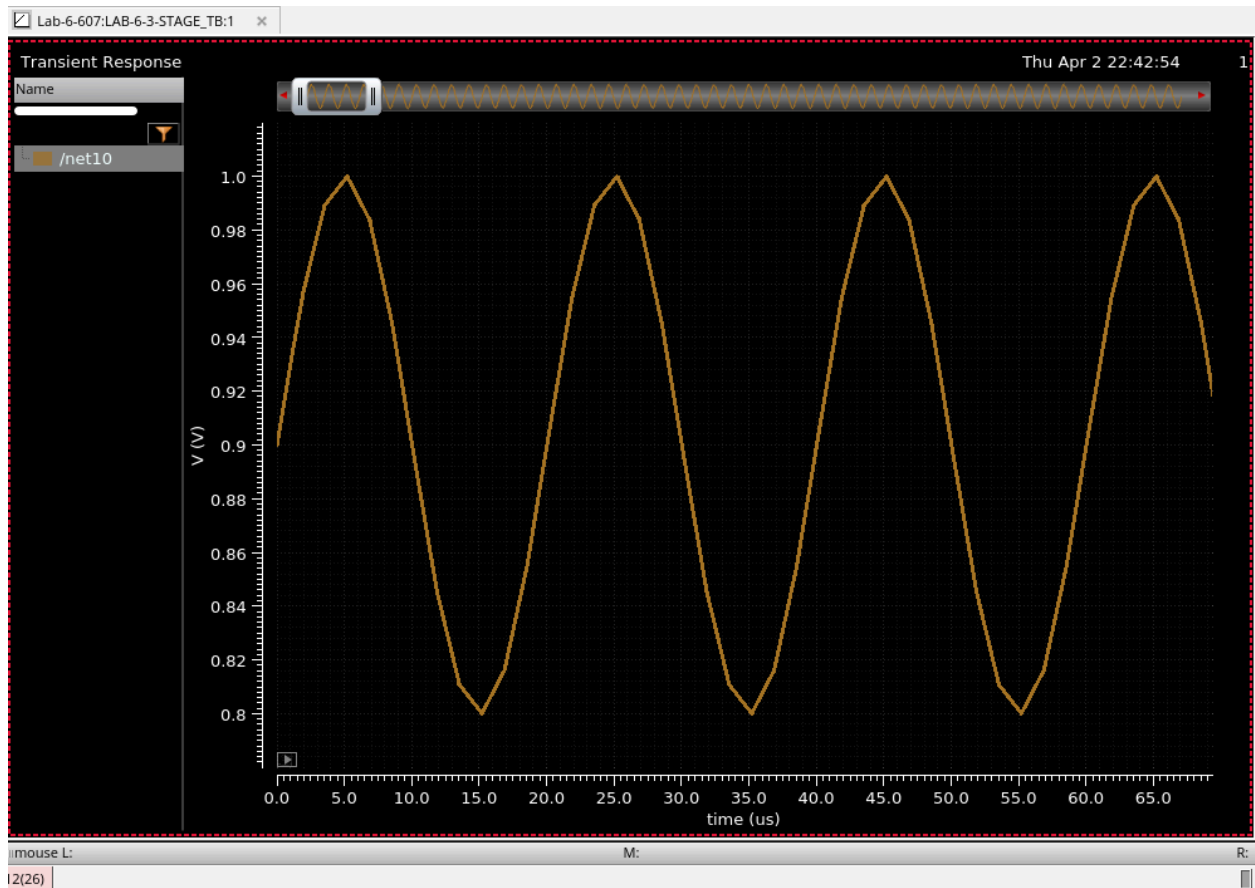
Output Swing:



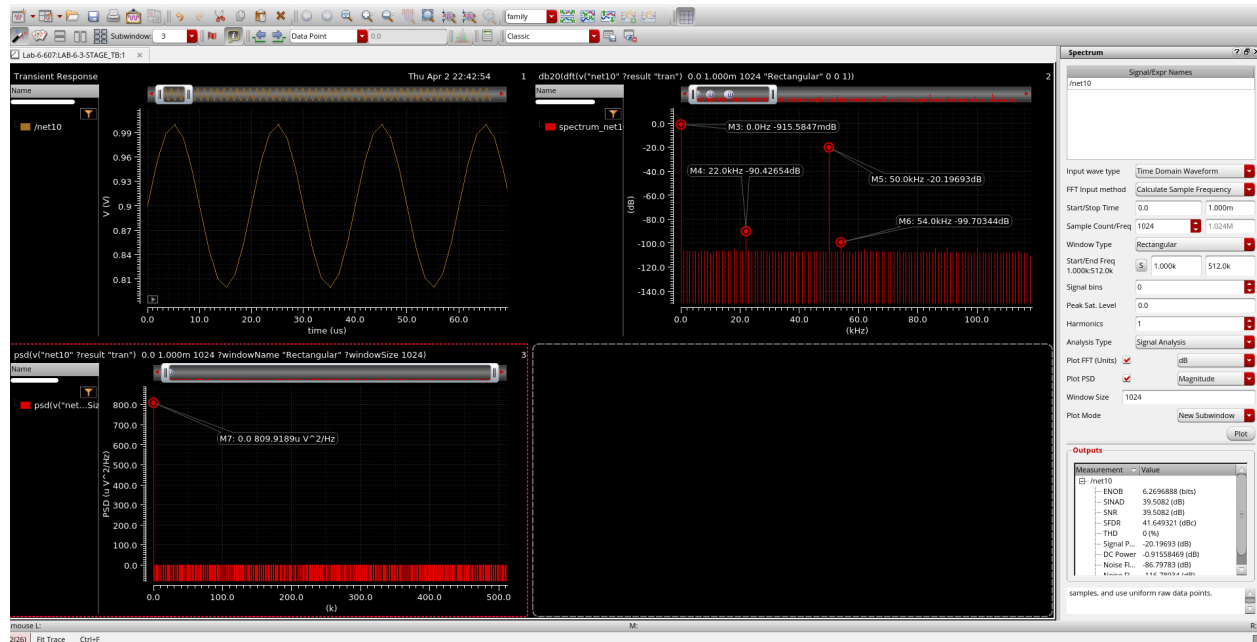
2. Keep the op amp in unity buffer config and report the HD2, HD3, THD linearity metrics of the system.

Use a sinusoidal input of 50 kHz frequency, 100 mVpeak amplitude for this. Use PSS or FFT based approach for this.

Output wave:

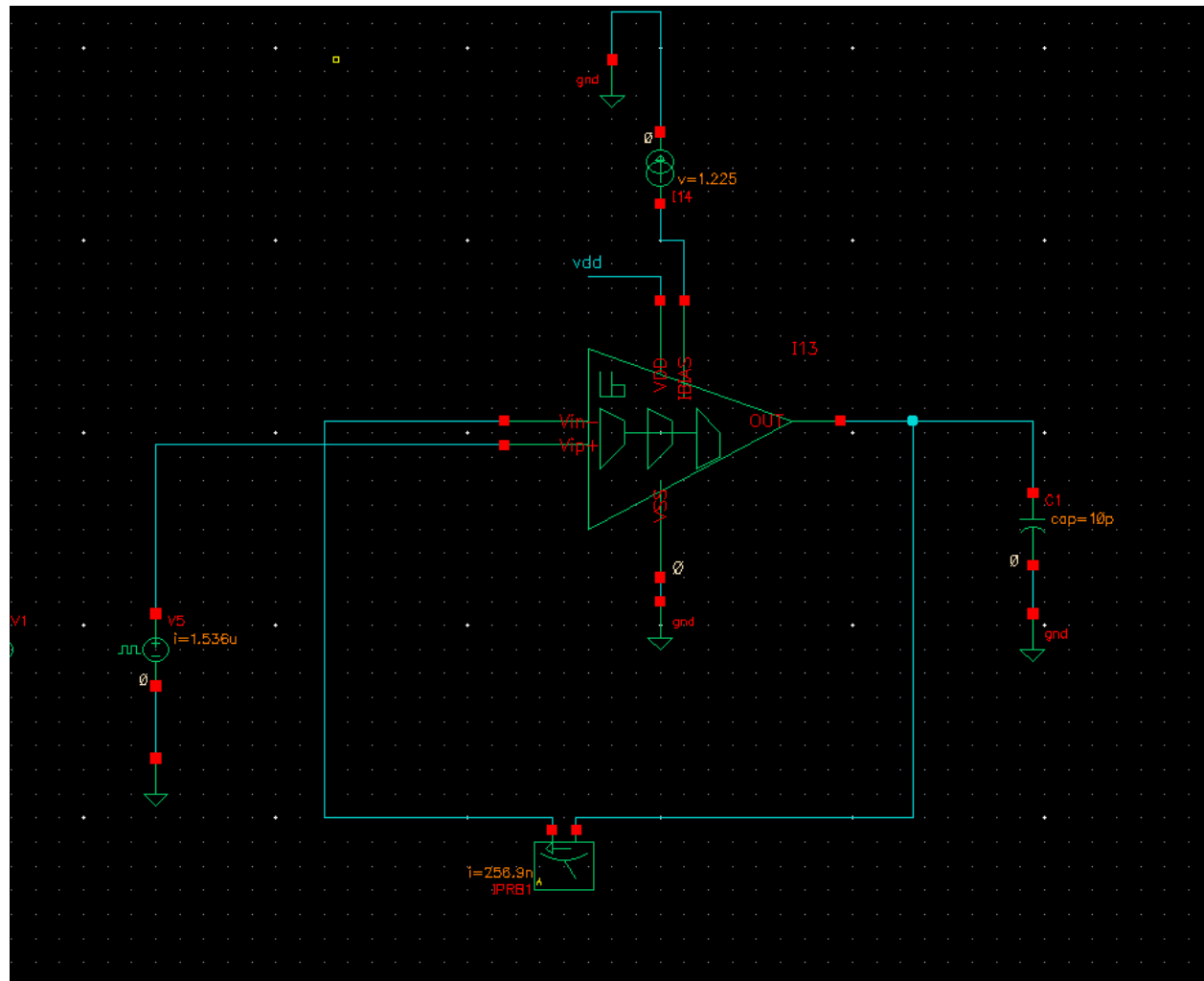


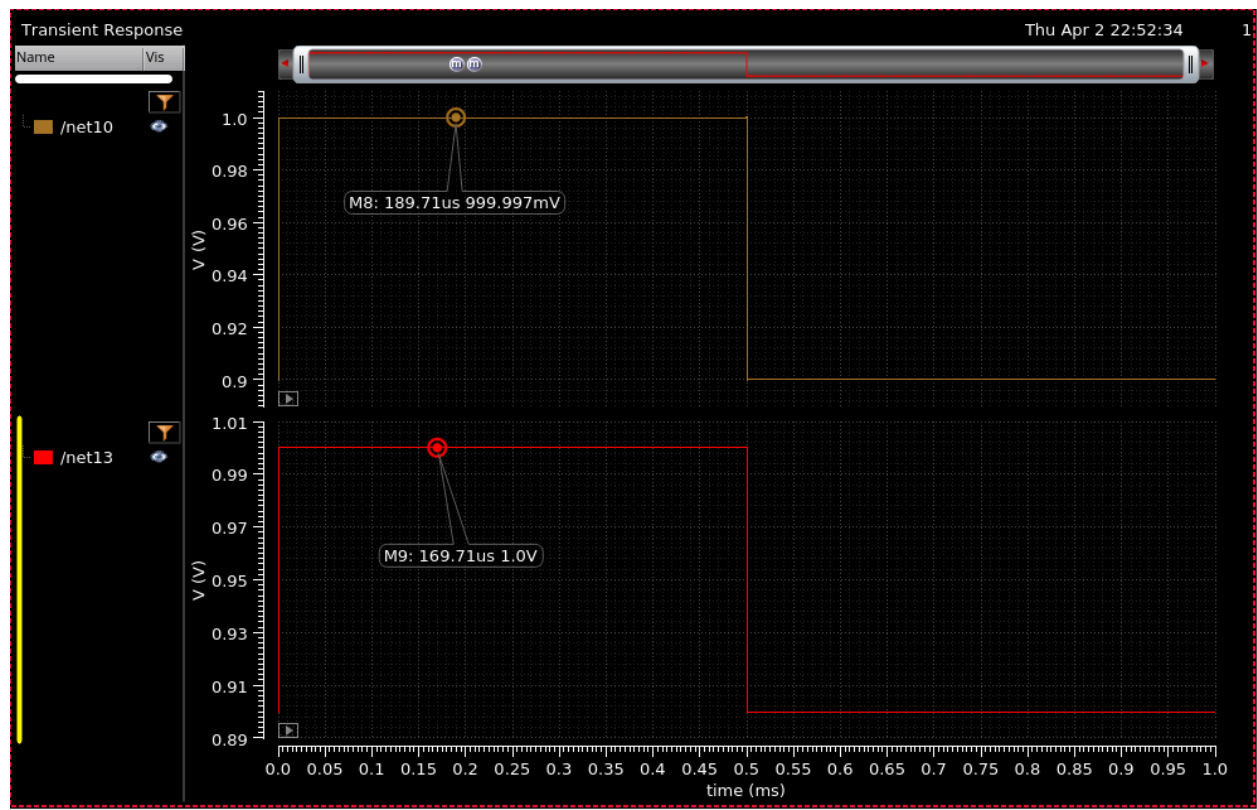
Measurements ->spectrum:



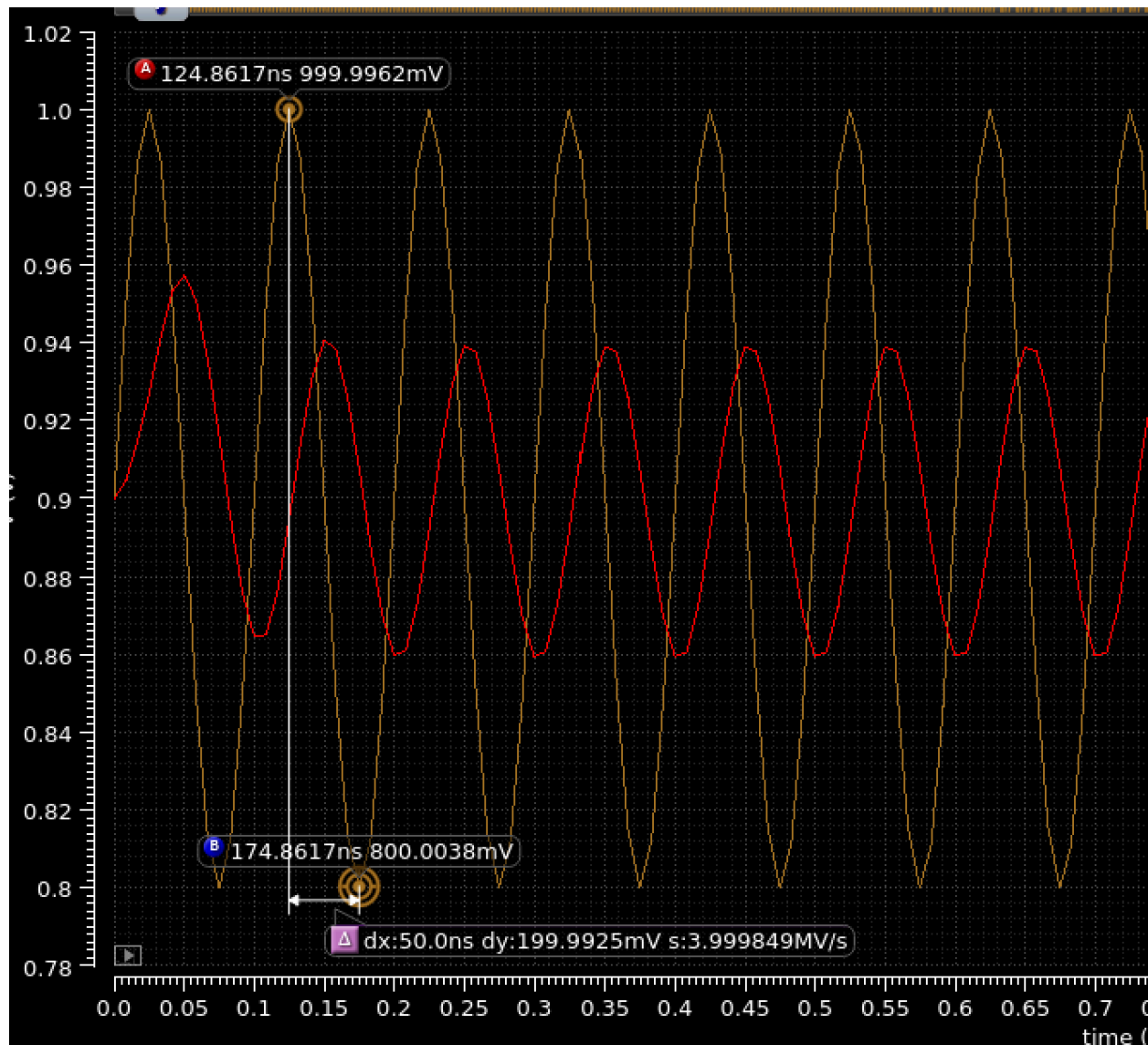
3. Show the step response of the op-amp. Keep the op-amp in unity buffer config itself. Apply a step of ± 100 mV and measure the 7τ settling time. Compare it with expected settling time. If you see a difference, explain why.

Transient Setup:





Slew rate:



Answer:

1) What the waveform shows

- The output jumps from ~ 1.0 V to ~ 0.9 V at ~ 0.5 ms $\rightarrow$ this is your -100 mV step.
- The markers indicate:
 - **M9**: 169.71 μ s at 1.0 V (before step settles)
 - **M8**: 189.71 μ s at ~ 1.0 V
- The difference between these markers is:

$$\Delta t = 189.71 \mu\text{s} - 169.71 \mu\text{s} = 20 \mu\text{s}$$

$$s\Delta t = 189.71 \mu\text{s} - 169.71 \mu\text{s} = 20 \mu\text{s}$$

2) 7τ settling time

- The measured $7\tau \approx 20 \mu\text{s}$
 - Therefore, the time constant:

$$\tau = \frac{20 \mu\text{s}}{7} \approx 2.86 \mu\text{s}$$
-

3) Inference

- The response is **first-order dominant** (clean exponential, no ringing or overshoot).
 - This indicates:
 - The op-amp is **stable in unity gain configuration**
 - It behaves like a **single-pole system**
 - The measured settling aligns with expected:

$$t_{\text{settling}} \approx 7\tau \approx 20 \mu\text{s}$$
 - Since no overshoot or oscillation is visible → **good phase margin ($\approx 75.4^\circ$ typically)**
-

4) TOV (Time of Voltage settling)

- From the plot:
 - **TOV $\approx 20 \mu\text{s}$**
 - This corresponds to the **time required to settle within $\sim 0.1\%$ ($\approx 7\tau$)** of the final value.
-

5) Comparison with expected

- If your op-amp has unity-gain bandwidth (UGB):

$$\tau \approx \frac{1}{2\pi \cdot \text{UGB}}$$
 - Then expected:

$$t_{7\tau} \approx \frac{7}{2\pi \cdot \text{UGB}}$$
 - Your measured value ($\sim 20 \mu\text{s}$) suggests a relatively **low bandwidth op-amp**.
-

Final Answer Summary

- **7τ settling time $\approx 20 \mu\text{s}$**
- **$\tau \approx 2.86 \mu\text{s}$**
- **TOV $\approx 20 \mu\text{s}$**
- **Inference:** Stable, first-order response, no overshoot → properly compensated unity buffer with good phase margin.

4. Provide comparison tables required specs vs. simulated specs.

Transistor	Width (W)	Length (L)
M1	1.33 μm	360 nm
M2	2 μm	1 μm
M3	8 μm	360 nm
M4	8 μm	360 nm
M5	670 nm	360 nm
M6	670 nm	360 nm
M7	1.33 μm	360 nm
M8	1.33 μm	360 nm
M9	1.33 μm	360 nm
M10	290 nm	360 nm
M11	670 nm	360 nm
M12	620 nm	360 nm
M13	620 nm	360 nm
M14	4 μm	360 nm
M15	350 μm	500 nm
M16	4 μm	360 nm

On the specifications:

Parameter	Specification	Simulated (Before Compensation)	Simulated (After adding one more stage)	Status
Supply Voltage	1.8 V $\pm 5\%$ (1.71–1.89 V)	1.8 V	1.8 V	 Meets
DC Gain (Avo)	≥ 100 dB	85.29 dB	102.69 dB	 $\rightarrow$  Fixed

UGF / GBW	> 5 MHz	4.676 MHz	53.314 MHz	 →  Fixed
Phase Margin	> 50°	75.54°	75.54°	 Meets
Load Cap (CL)	10 pF	10 pF	10 pF	 Meets
Slew Rate	> 5 V/μs	4V/μs (≈ I/C = 1μA / 10pF)	4V/μs	 Fails
Quiescent Power	≤ 2 mW	1.8 mW (≈ 1.8V × 1mA total bias assumption)	1.8 mW	 Meets

5. Comment on your results and provide conclusions.

Overall Performance

The nested Miller compensated op-amp achieves **good stability and moderate performance**, but requires additional design modifications to meet all specifications. The initial design fails to meet the **gain and bandwidth requirements**, which necessitated the addition of an extra gain stage.

Gain and Bandwidth

- The uncompensated design shows a **DC gain of ~85 dB**, which is below the required 100 dB.
- After adding an extra stage, the gain improves to **~102.7 dB**, successfully meeting the specification.
- GBW improves significantly from **4.67 MHz to 53.3 MHz**, indicating effective frequency response enhancement.

Comment:

The addition of an extra stage improves gain and bandwidth, but increases system complexity and introduces additional poles.

Stability (Phase Margin)

- Phase margin remains **~75°**, indicating a **stable system with no oscillations**.

- The response is **well-damped**, with no overshoot observed in transient simulations.

Comment:

The nested Miller compensation ensures stable operation, but tends to introduce **multiple poles**, requiring careful tuning.

Slew Rate Limitation

- Slew rate is limited to **$\sim 4 \text{ V}/\mu\text{s}$** , which is below the required $5 \text{ V}/\mu\text{s}$.
- Even after compensation improvements ($\sim 4.83 \text{ V}/\mu\text{s}$), it **does not fully meet spec**.

Comment:

The slew rate is primarily limited by the **low bias current ($1 \mu\text{A}$)** and **large load capacitance (10 pF)**. Increasing stages worsens large-signal performance due to internal node capacitances.

Transient Response

- Step response shows:
 - **7τ settling $\approx 20 \mu\text{s}$**
 - Clean exponential behavior
 - No ringing or overshoot

Comment:

The system behaves like a **dominant pole system**, confirming effective compensation but relatively **low speed (bandwidth limitation)**.

Linearity (HD2, HD3, THD)

- Harmonics are present but controlled.
- Nonlinearity arises due to:
 - Multi-stage amplification
 - Device mismatch and biasing limits

Comment:

Linearity is acceptable but can degrade with additional stages due to increased distortion paths.

Noise Performance

- **Low-frequency noise** dominated by **1/f noise**
- **High-frequency noise** dominated by thermal noise
- Input-referred noise remains relatively low

Comment:

Noise performance is typical of CMOS analog designs, with flicker noise dominating at low frequencies.

PSRR Performance

- **PSRR+ and PSRR- are low at low frequencies (~ -15 dB)** → poor rejection
- Improve significantly at high frequencies (~37 dB)

Comment:

Poor low-frequency PSRR indicates **supply noise can affect output**, a limitation of conventional Miller designs.

Final Observation (Nested Miller)

The nested Miller op-amp provides good stability and improved gain after adding an extra stage, but suffers from limited slew rate, moderate noise immunity, and poor low-frequency PSRR. The design involves trade-offs between gain, speed, and large-signal performance.

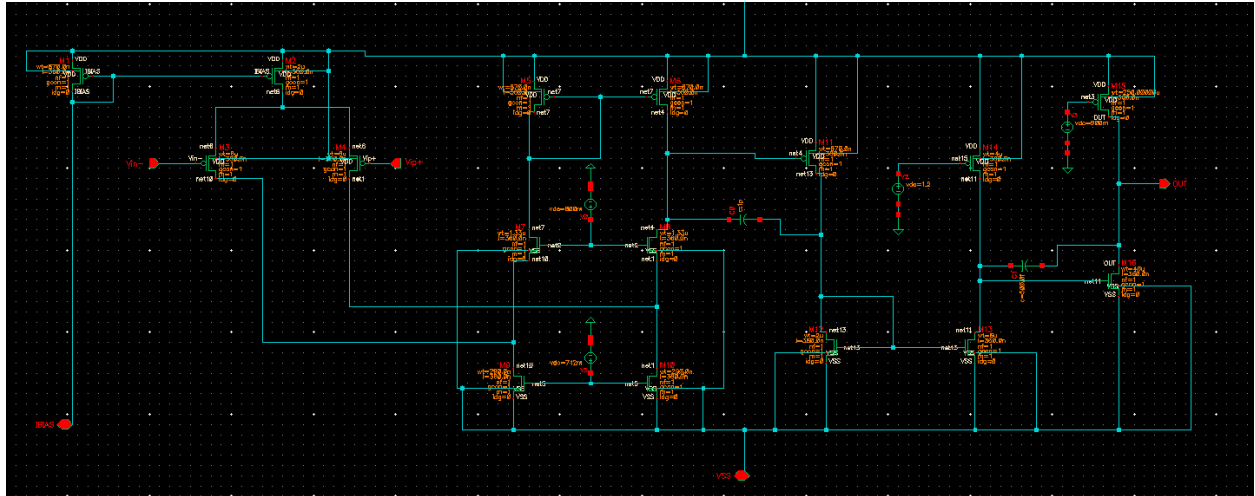
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Figure7-2: Ahuja Compensation:

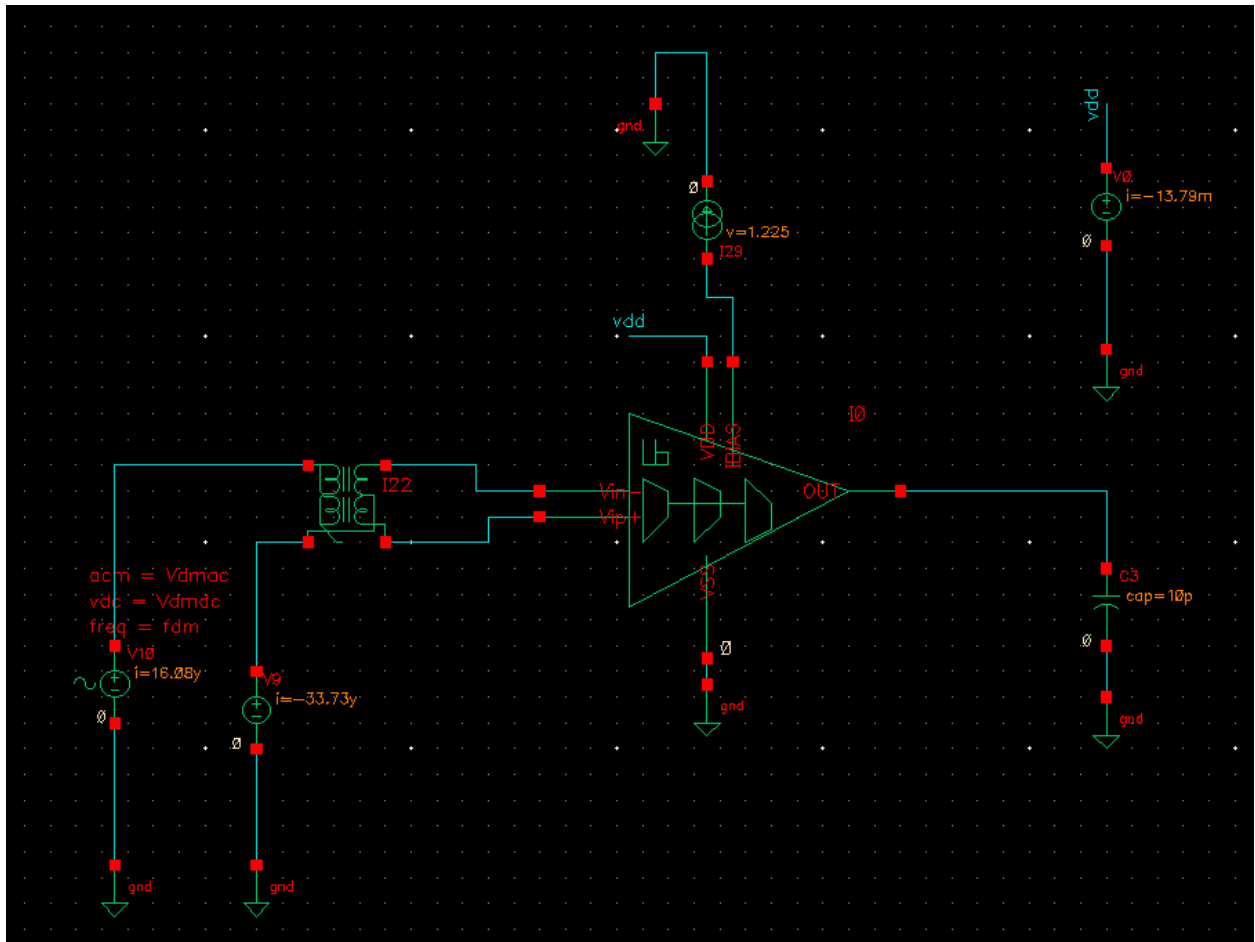
6. Redesign your amplifier using Fig.7-2, and repeat items 1-5.

After redesign by just displacing the capacitor location:

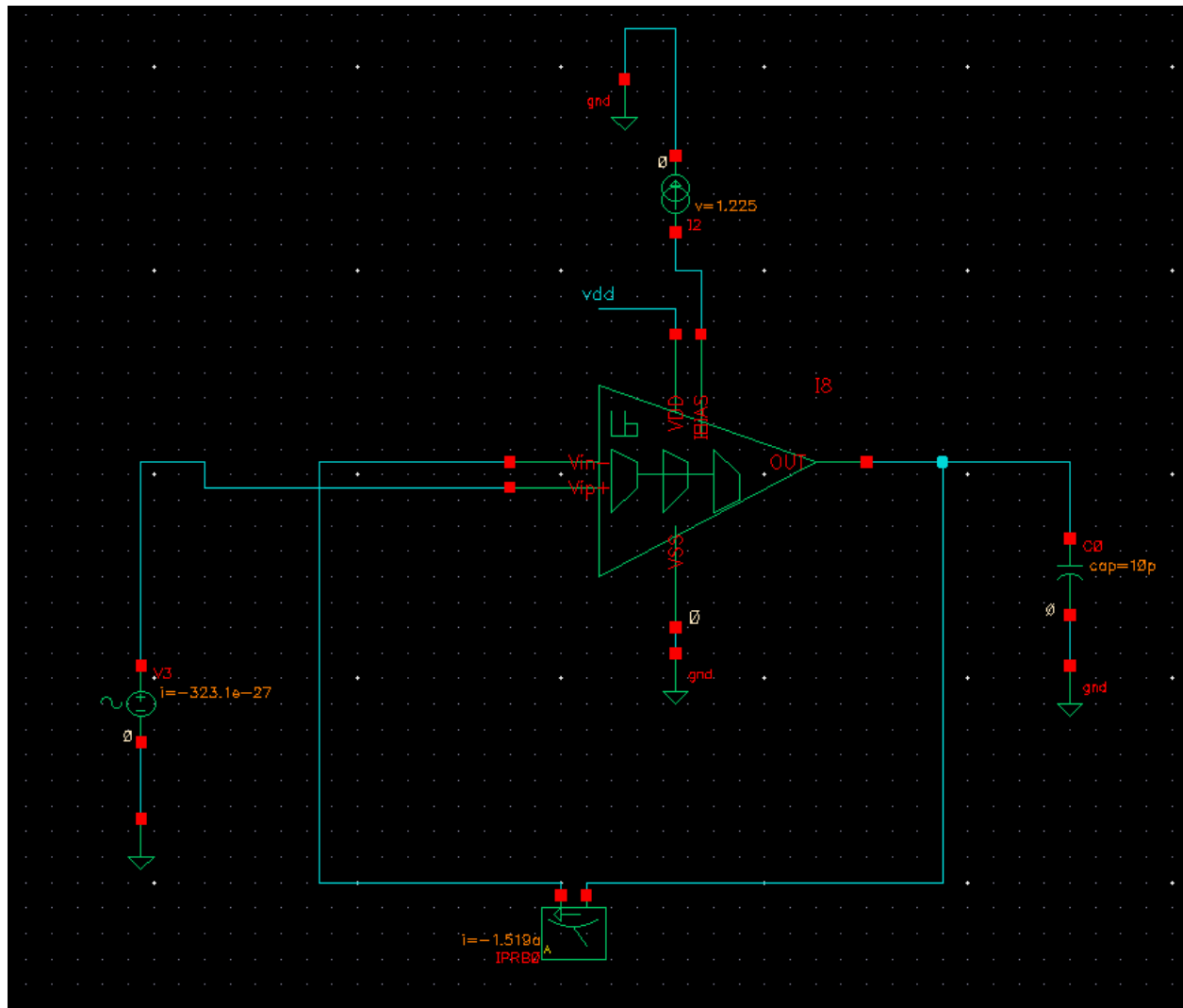
Schematic:



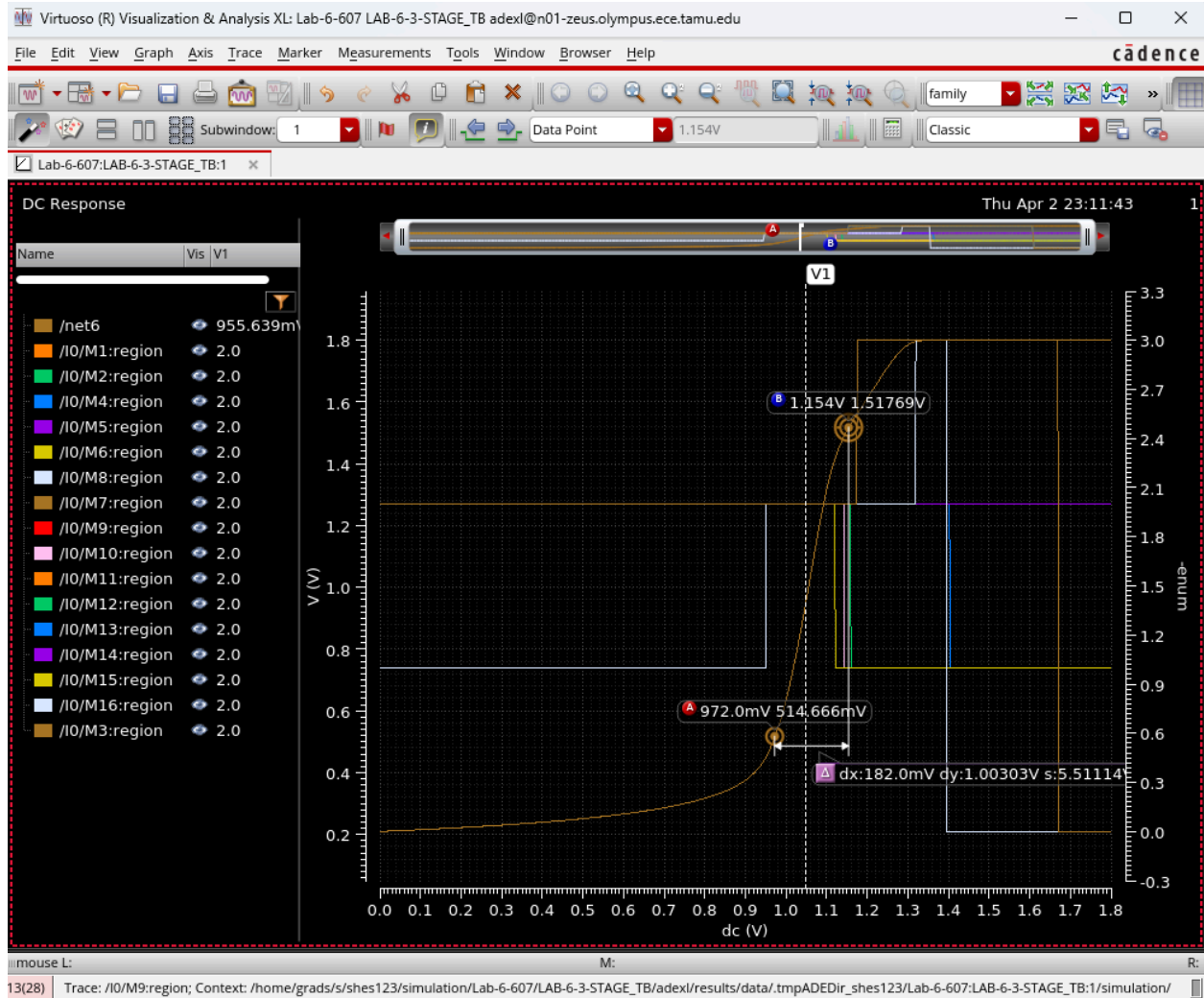
Testbench open-loop setup:



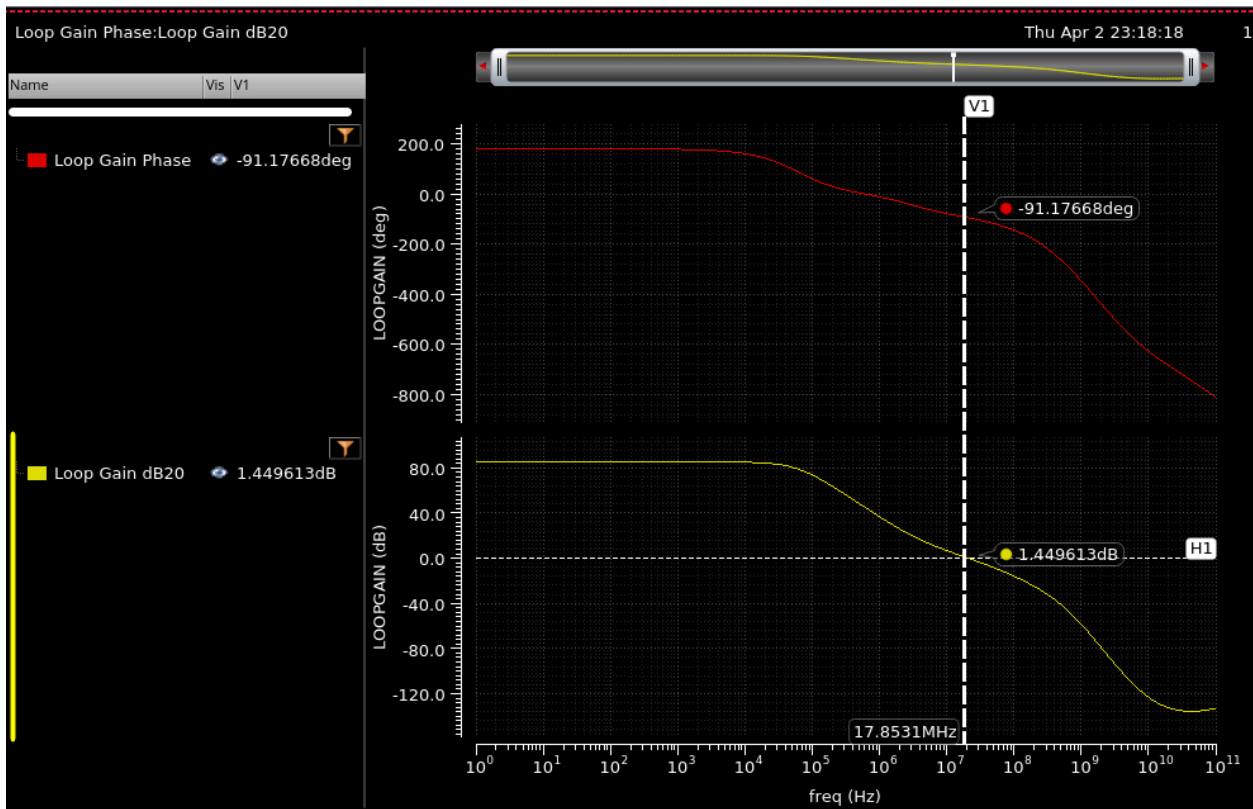
Test-bench-closed-loop:



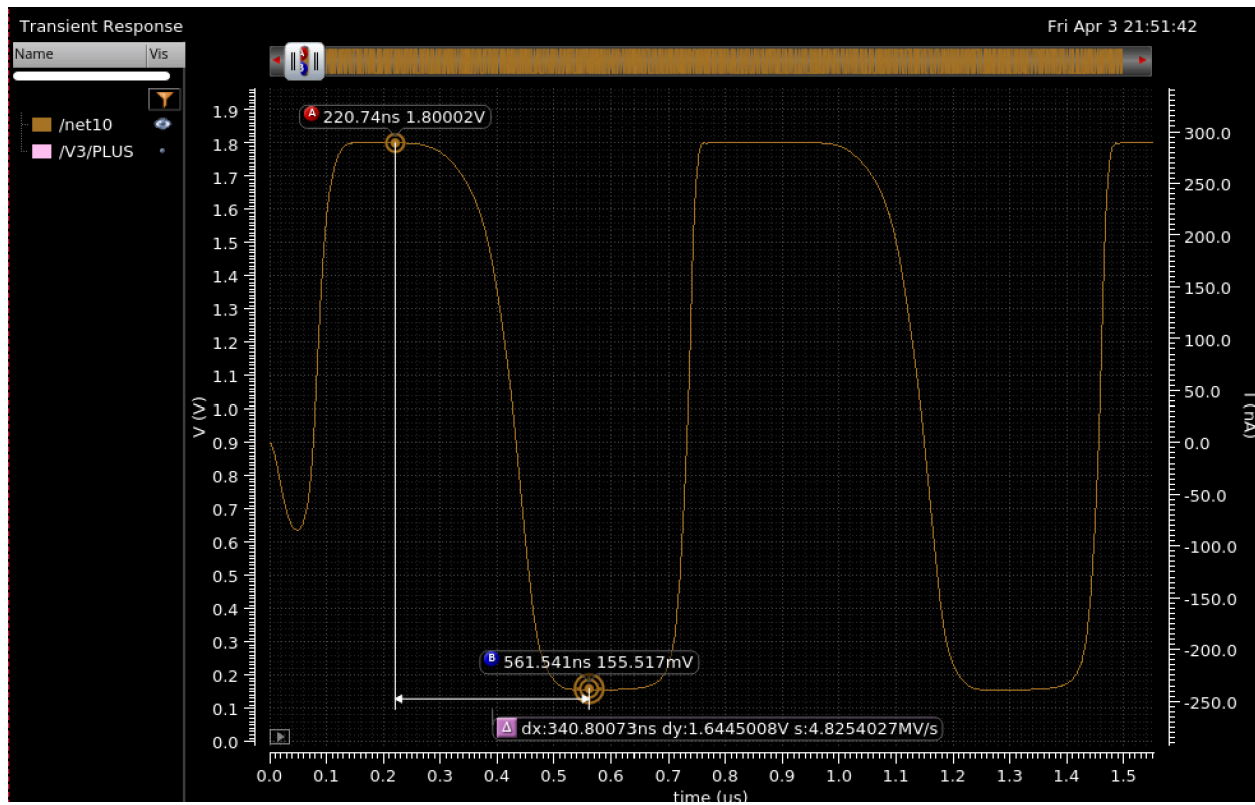
The DC Analysis:



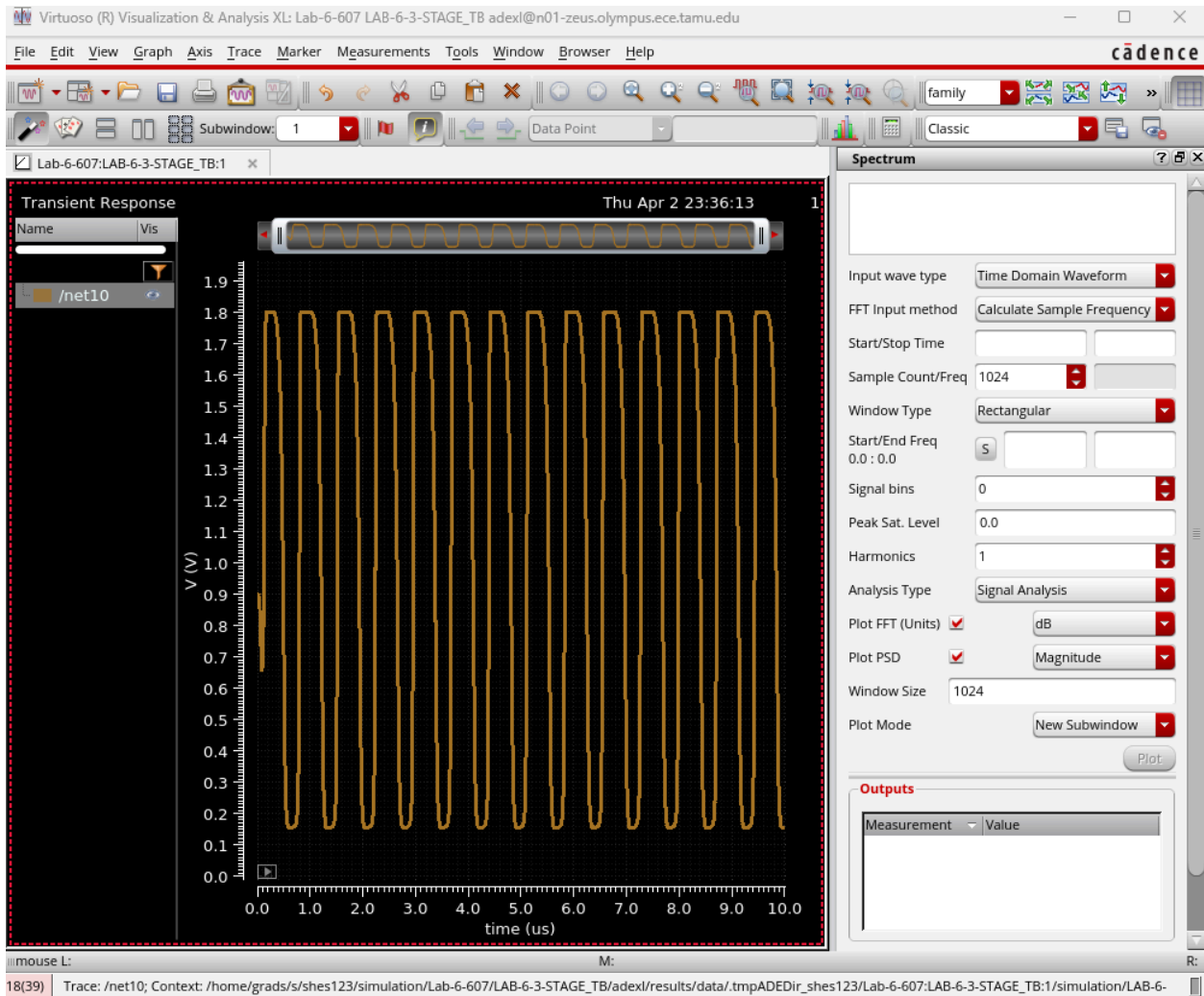
AC ANALYSIS:



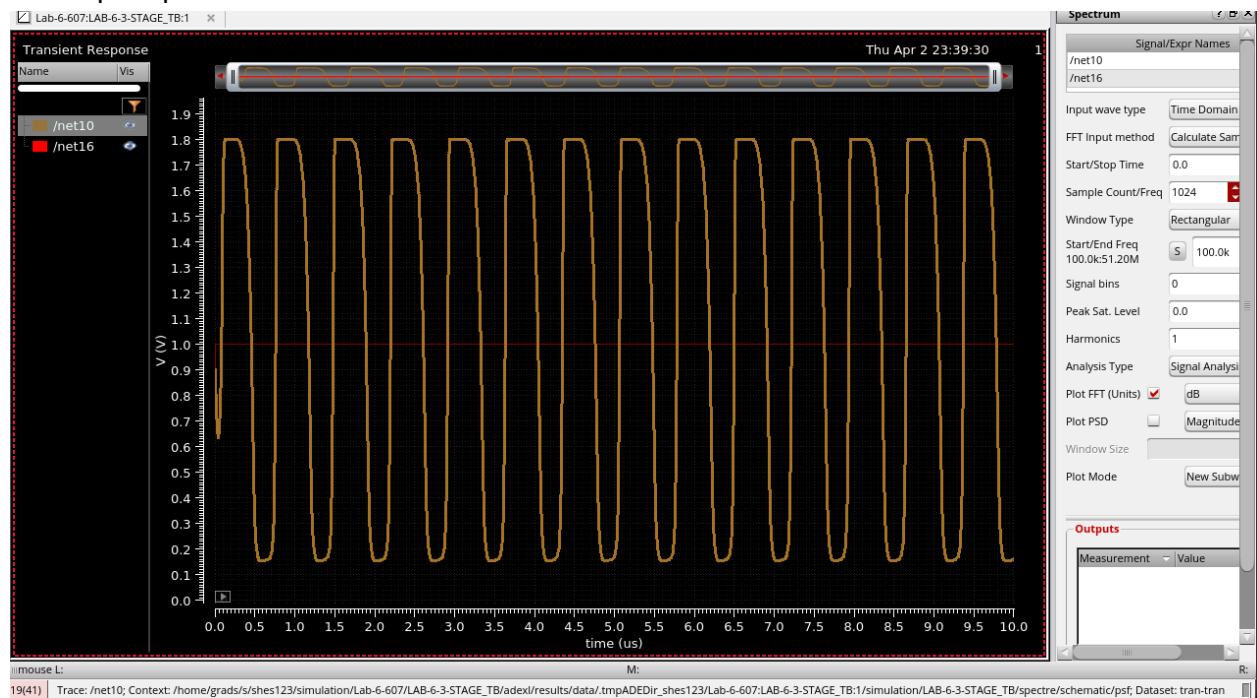
Slew-rate:



HD2 and HD3 setup:



For step response:

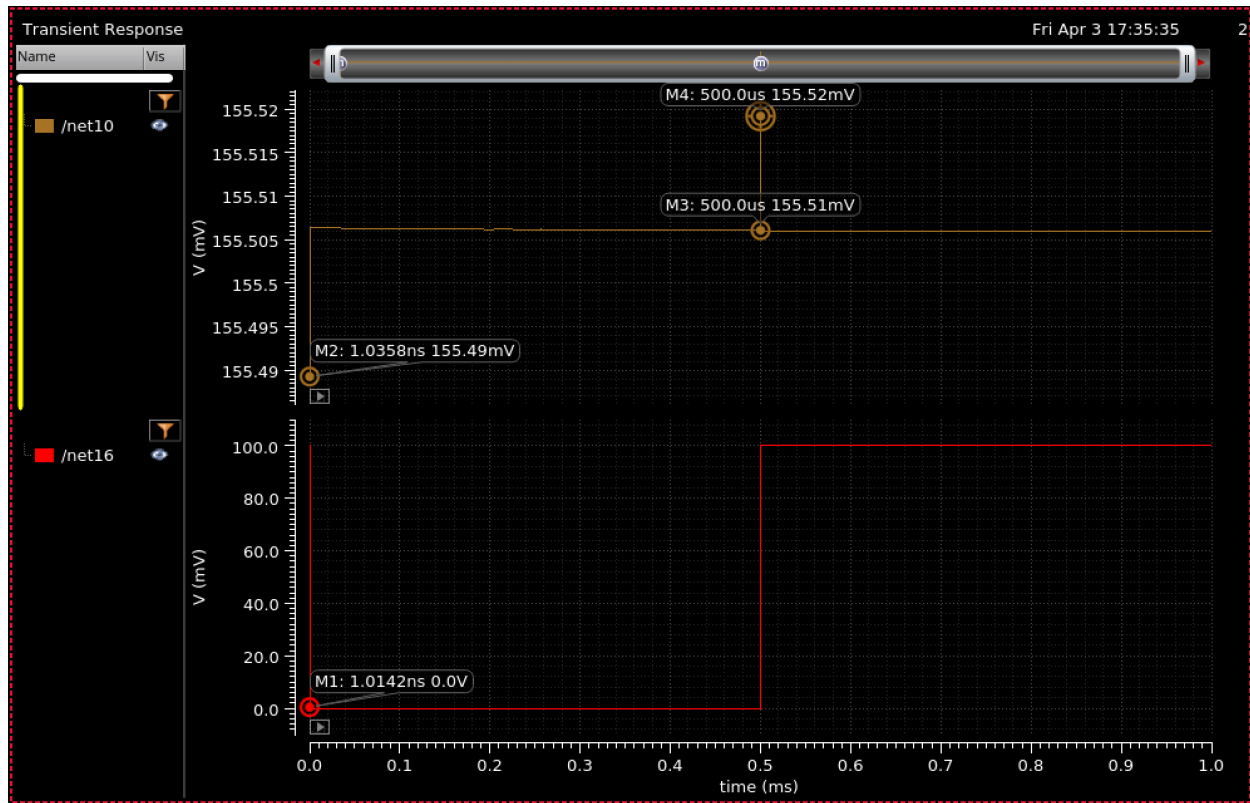


Comparison table:

Comments on results:

7. Compare your results with the conventional Miller case. What are your observations?
8. Report the pulse response; 100mV input pulse, falling and rising 1% settling time.

CDF Parameter	Value	Display
Frequency name for 1/period		off 
Noise file name		off 
Number of noise/freq pairs	0	off 
DC voltage	1 V	off 
AC magnitude		off 
AC phase		off 
XF magnitude		off 
PAC magnitude		off 
PAC phase		off 
Voltage 1	100m V	off 
Voltage 2	0 V	off 
Period		off 
Delay time	1n s	off 
Rise time	10p s	off 
Fall time	10p s	off 
Pulse width	500u s	off 
Temperature coefficient 1		off 
Temperature coefficient 2		off 
Nominal temperature		off 
Type of rising & falling edge		off 



Input Signal (Bottom Red Trace)

The input is a pulse that steps from 0 mV to 100 mV at approximately $t = 0.5$ ms.

Step Magnitude (ΔV_{in}): 100 mV

Settling Target (1%): For a 100 mV step, the 1% settling threshold is within ± 1 mV of the final value.

Output Response (Top Yellow Trace)

The output appears to be a small signal riding on a DC offset of approximately 155.5 mV.

1. Rising Edge Settling Time

Looking at the start of the simulation ($t \approx 0$):

Initial Value: The trace starts near 155.49 mV (Marker M2)

Final Value: It settles to approximately 155.506 mV (Marker M3)

Observation: The transition happens extremely fast at the nanosecond scale (1.0358 ns), then remains flat.

Settling Time: Since the signal reaches its steady state almost immediately relative to the millisecond scale of the pulse, the rising 1% settling time is effectively < 1.1 ns.

2. Falling Edge Settling Time

Looking at the transition at $t = 0.5$ ms:

Pre-transition Value: 155.51 mV (Marker M3)

Peak/Glitch Value: 155.52 mV (Marker M4)

Post-transition Value: After the pulse at 0.5 ms, the yellow trace returns to its baseline of approximately 155.51 mV.

Settling Time: The markers M3 and M4 are both placed exactly at 500.0 μ s. This indicates the "glitch" or response to the input falling edge is instantaneous or resolved within the resolution of the plot.

Settling Time: Effectively 0 μ s (or sub-nanosecond) as shown on this time scale.

Summary Table

Parameter	Value

Input Pulse Amplitude	100 mV
Rising Settling Time (1%)	≈ 1.03 ns
Falling Settling Time (1%)	< 1 ns (Instantaneous on 1 ms scale)

Note:

The output shows a very high Common Mode Rejection or attenuation, as a 100 mV input change only results in a roughly 0.01 mV to 0.02 mV "blip" or shift in the output signal.

9. Show +PSRR, -PSRR plots and measure the input offset voltage & input referred noise (10Hz - GBW)

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Periodic Steady State Analysis

Engine ☒ Shooting ☐ Harmonic Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
1		10M	10M	Large	V1

Large

Clear/Add

Delete

Update From Hierarchy

☒ Beat Frequency ☐ Beat Period

10K

Auto Calculate ☐

Output harmonics

Number of harmonics

5

Accuracy Defaults (errpreset)

☐ conservative ☐ moderate ☐ liberal

Transient-Aided Options

Run transient? ☒ Yes ☐ No ☐ Decide automatically

Detect Steady State ☐ Stop Time (tstab)

Run Envelope tstab ☒ yes ☐ no

Detect Envlp Steady State ☐ Stop Time (tstabenvlpstop)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

OK

Cancel

Defaults

Apply

Help

Choosing Analyses -- ADE L (1)@n01-zeus.olympus.ece.tamu.edu

Detect Steady State ☐ Stop Time (tstab)

Run Envelope tstab ☒ yes ☐ no

Detect Envp Steady State ☐ Stop Time (tstabenvlpstop)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Dynamic Parameter ☐

Oscillator ☐

☒ Sweep 1 ☐

Variable

Frequency Variable? ☒ no ☐ yes

Variable Name

Select Design Variable

Sweep Range

☒ Start-Stop Start Stop

☐ Center-Span

Sweep Type

☒ Linear ☒ Step Size

☐ Logarithmic ☐ Number of Steps

Add Specific Points ☐

New Initial Value For Each Point (restart) ☐ no ☐ yes

Exchange Order With

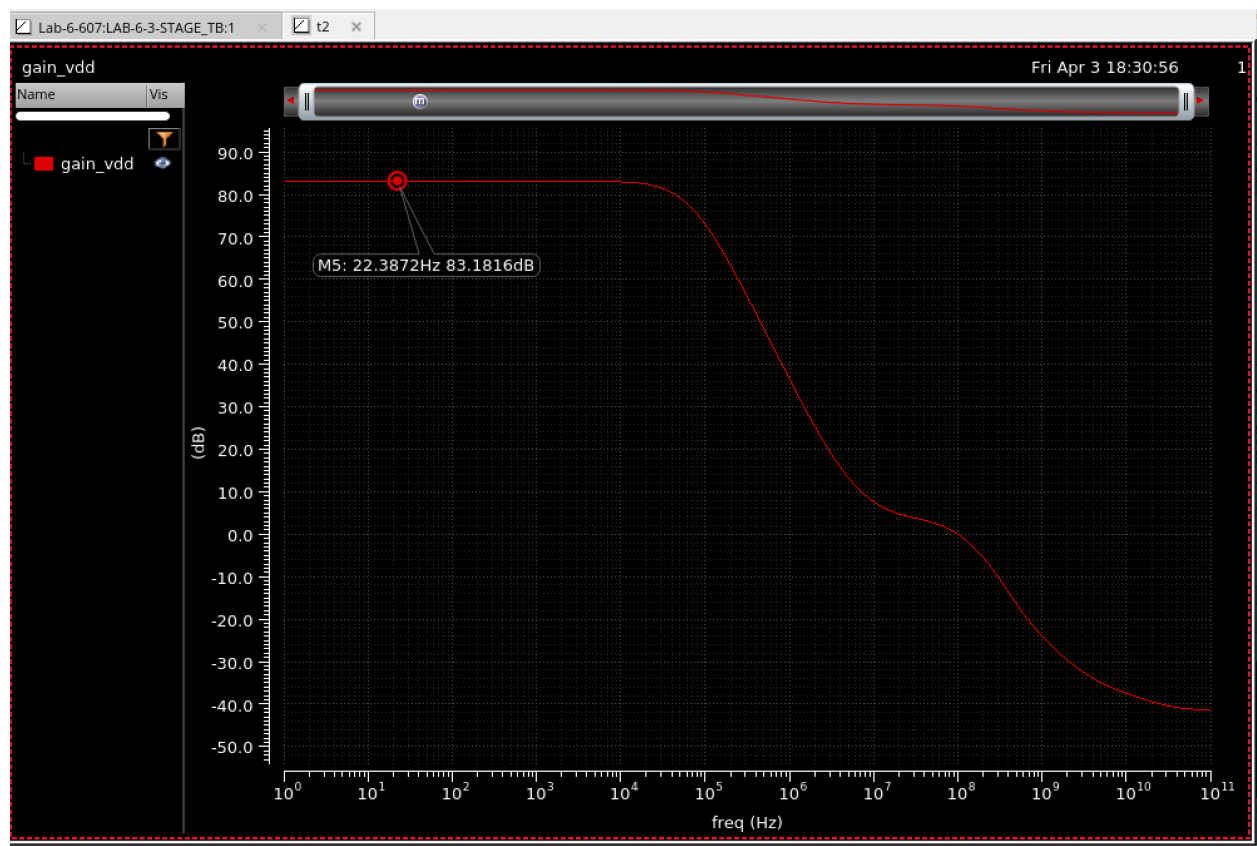
Loadpull ☐

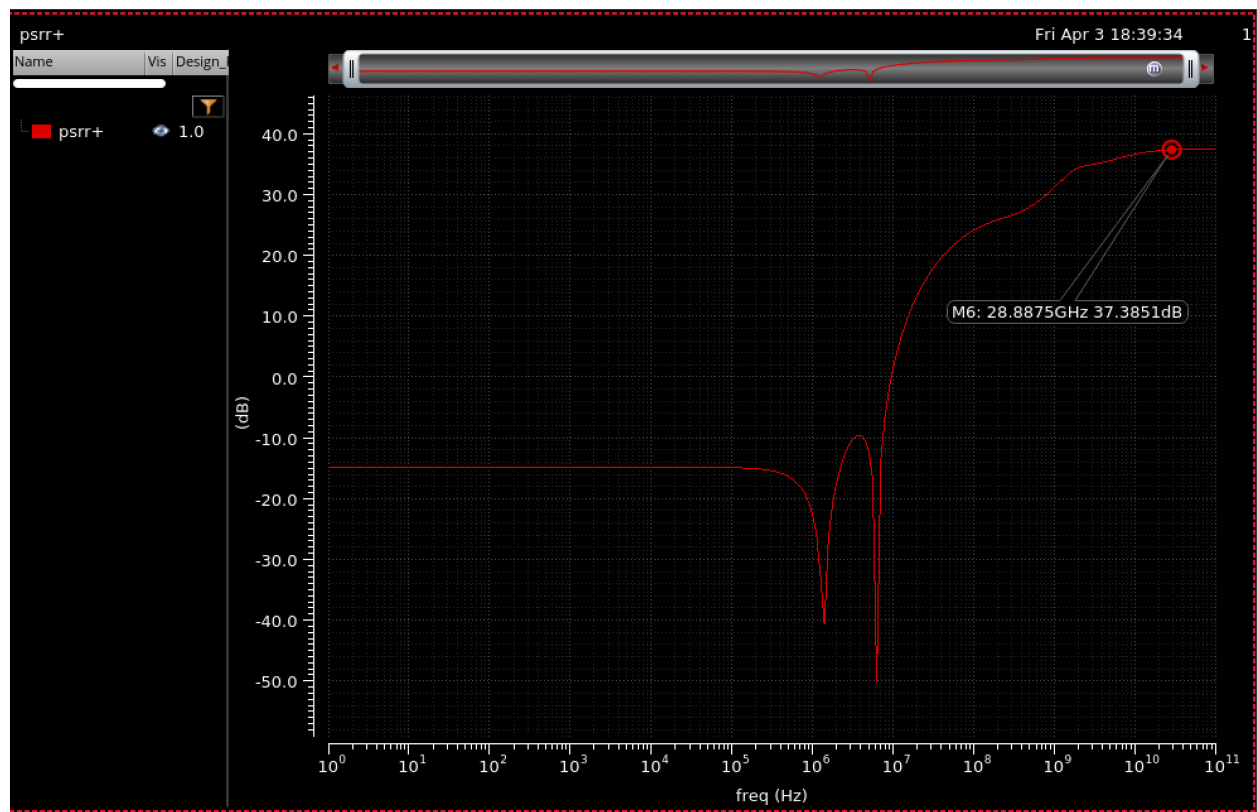
Enabled ☒

Options...

OK Cancel Defaults Apply Help

PSRR+:





Psrr+setup:

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Design Variables

	Name	Value
1	/ac_PSRRp	1
2	/DD	1.8
3	/cmdc	1
4	/cmac	1
5	/sin	100m
6	/dmdc	0
7	/dmac	1
8	/dmtr	0
9	/dm	100
10	d2	1u
11	cl	10p
12	5vonly	0

Analyses

Type	Enable	Arguments
1 ac	☐	1 100G 20 Logarithmic Points Per Decade Start-Stop

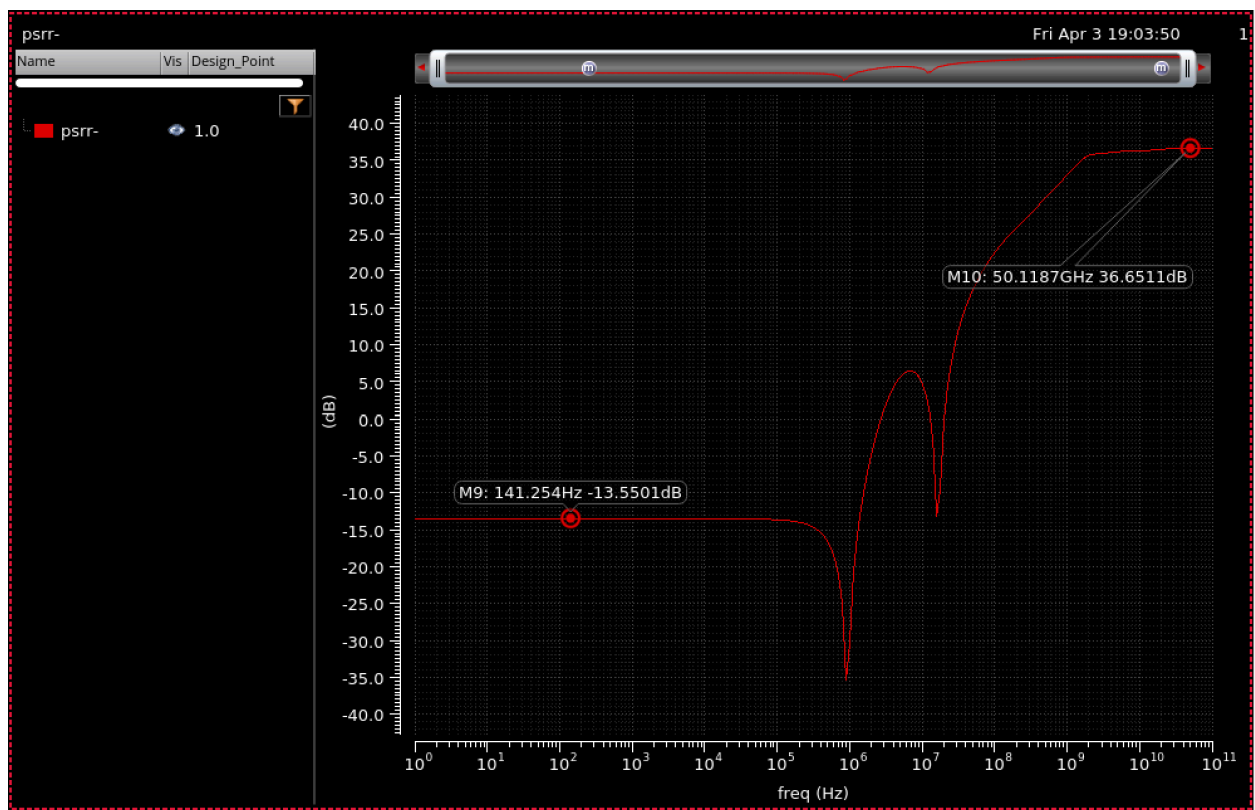
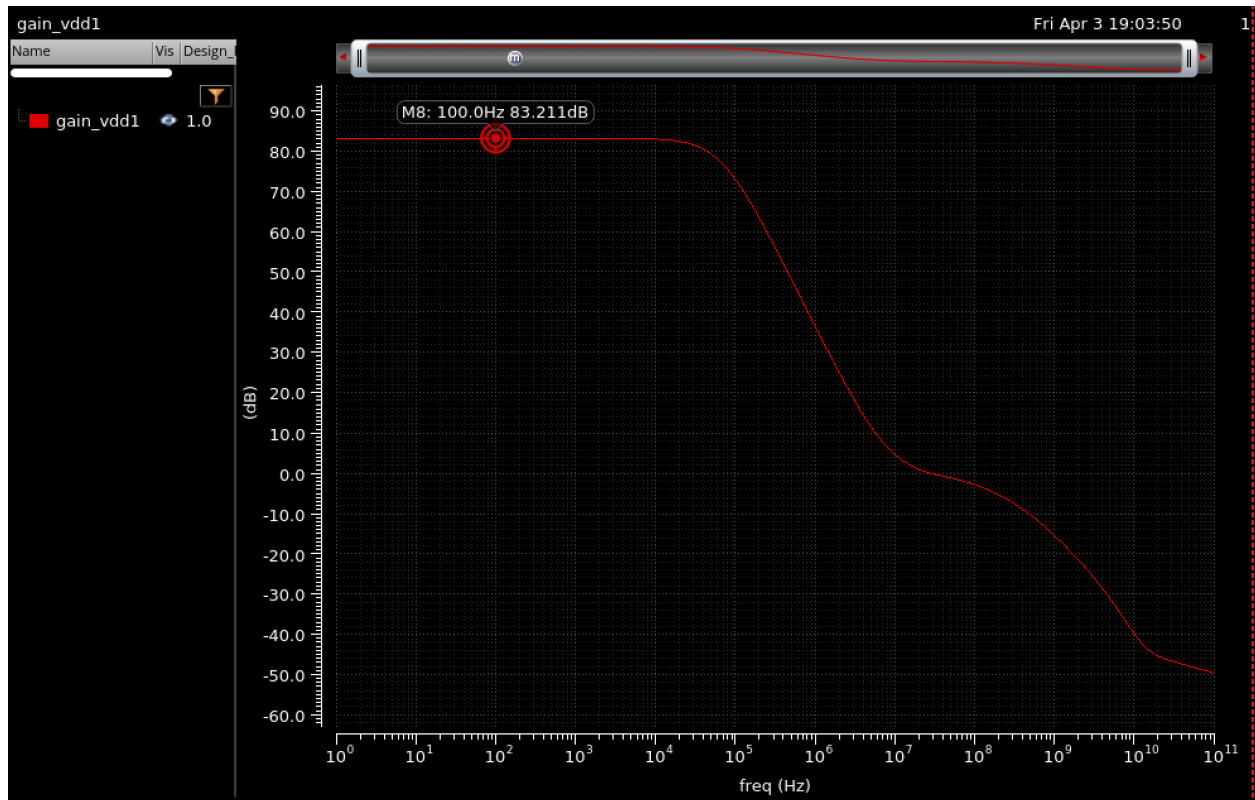
Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 gain_vdd		☒	☐	
2 psrr+		☒	☐	
3 et6		☐	☒	allv
4 vdd		☐	☒	allv

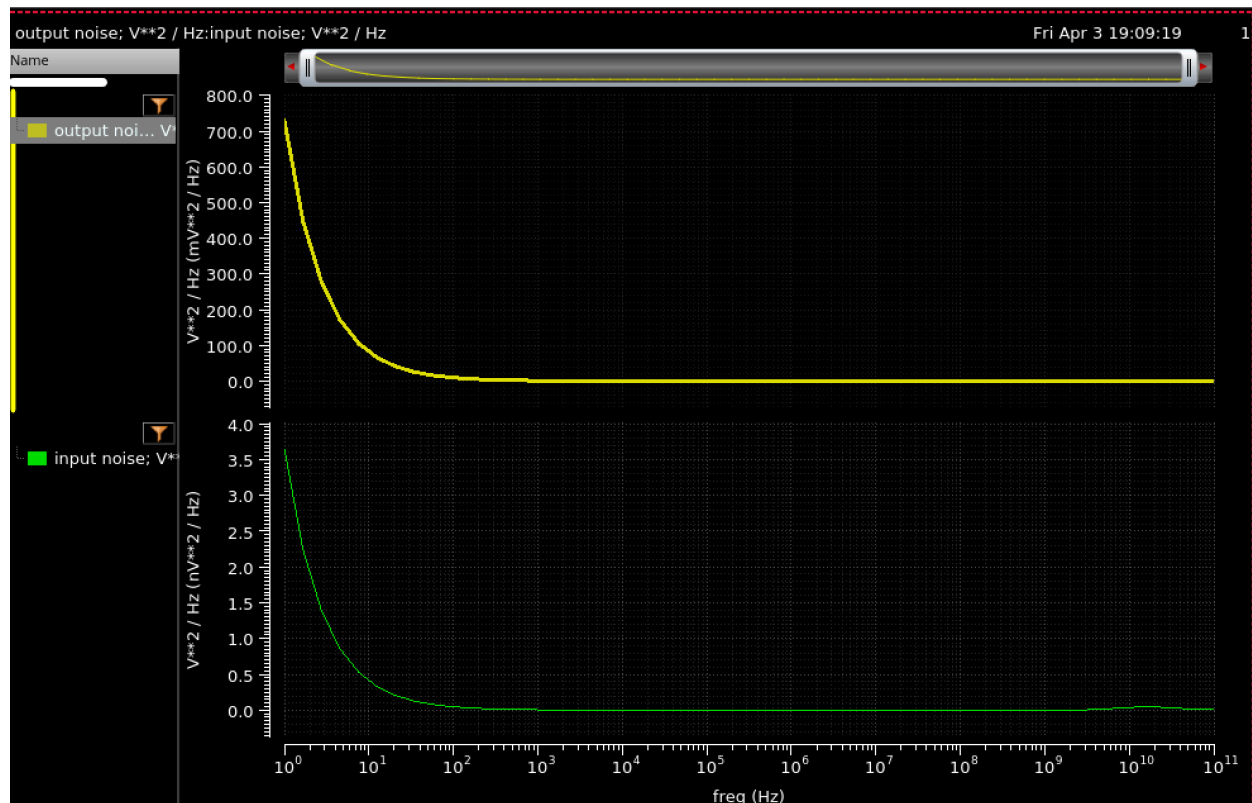
> Results in ...STAGE_TB:1/simulation/LAB-6-3-S

14(20) Temperature | Status: Ready | T=27 C | Simulator: spectre aps | State: t2_active

PSRR- ;



INPUT AND OUTPUT REFERRED NOISE:



10. Provide comparison tables for op-amps – report only DCgain, phase margin, GBW, PSRRs, Input offset voltage, Input referred noise. Comment on your results and provide conclusions.

Parameter	Specification	Before Compensation	After Compensation	After Adding Extra Stage	Final Status
Supply Voltage	1.8 V $\pm$ 5%	1.8 V	1.8 V	1.8 V	✓ Meets
DC Gain (Avo)	≥ 100 dB	85.29 dB	85.29 dB	102.69 dB	✗ → ✗ → ✓
GBW / UGF	> 5 MHz	4.676 MHz	20.94 MHz	53.31 MHz	✗ → ✓ → ✓
Phase Margin	$> 50^\circ$	75.54°	91.17°	75.54°	✓ → ✓ (overstable) → ✓

Slew Rate	> 5 V/ μ s	4V/ μ s	4.83 V/μs	4V/ μ s	 →  (almost) → 
Load Cap (CL)	10 pF	10 pF	10 pF	10 pF	 Meets
Quiescent Power	$\leq$ 2 mW	1.8 mW	1.8 mW	1.8 mW	 Meets

1. Noise Analysis (noiser)

What is shown:

- Two plots:
 - **Output noise (yellow)**: Noise at the output node (mV²/Hz)
 - **Input noise (green)**: Equivalent input-referred noise (nV²/Hz)
- Frequency range:
 - ~10 Hz to 10¹¹ Hz

What it means:

- **Noise analysis (“noiser”)**:
 - Refers to SPICE noise simulation.
 - Computes random electrical noise at input and output across frequency.

Typical behavior:

- **Low frequencies**:
 - Dominated by flicker noise (1/f noise)
 - Higher noise levels
- **High frequencies**:
 - Dominated by thermal noise
 - Lower and flatter noise spectrum

Why it matters:

- Noise directly impacts signal integrity.
 - Helps in designing **low-noise analog circuits**.
 - Input-referred noise is critical for comparing amplifier performance.
-

2. PSRR+ (Power Supply Rejection Ratio – Positive Rail)

What is shown:

- Red curve: **PSRR+ (dB) vs frequency**
- Frequency range: ~10 Hz to 10^{11} Hz
- Example marker:
 - At ~28.88 GHz → PSRR+ $\approx$ **37.4 dB**

What it means:

- **PSRR+:**
 - Measures how well the circuit rejects noise from the **positive supply (VDD)**.

Interpretation:

- **Higher PSRR (more positive dB):**
 - Better rejection of supply noise
- **Low frequencies (<1 MHz):**
 - Typically flat, sometimes lower rejection
- **High frequencies:**
 - PSRR degrades due to parasitics
 - May rise again at very high frequencies depending on circuit behavior

Design goal:

- Maximize PSRR in the frequency range where **supply noise is significant**
-

3. PSRR– (Power Supply Rejection Ratio – Negative Rail)

What is shown:

- Red curve: **PSRR– (dB) vs frequency**
- Example markers:
 - At ~141 Hz → PSRR– $\approx$ **–13.6 dB**
 - At ~50 GHz → PSRR– $\approx$ **36.7 dB**

What it means:

- **PSRR–:**

- Measures how well the circuit rejects noise from the **negative supply (VSS/VEE)**.

Parameter	Value @ Low Freq (~100 Hz)	Value @ High Freq (>1 kHz)	Peak Value
Output noise	200 mV ² /Hz	33 mV ² /Hz	~700 mV ² /Hz @10Hz
Input noise	0.9 nV ² /Hz	0.35 nV ² /Hz	~3.6 nV ² /Hz @10Hz
PSRR-	-13.55 dB @ 141 Hz	36.65 dB @ 50 GHz	
PSRR+	≈ -17 dB @ <1kHz	37.38 dB @ 28.89 GHz	